

# Data Analysis Project Report: Statistical Business Analysis

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## Project Description

This project represents an advanced statistical business analysis pipeline built using Python. By leveraging the `scipy`, `statsmodels`, and `pandas` libraries, raw e-commerce and customer retention data are transformed into mathematically validated business insights. The analysis focuses on testing data distributions, generating confidence intervals, proving correlations, and executing rigorous hypothesis testing to empower data-driven decision-making.

## Objectives

- Utilize `scipy.stats` for advanced hypothesis testing (T-tests, ANOVA, Shapiro-Wilk).
- Generate exact p-values to mathematically prove or disprove business assumptions.
- Calculate 95% Confidence Intervals to determine exact margins of error for revenue projections.
- Build an Ordinary Least Squares (OLS) Linear Regression model to quantify variance.
- Consolidate statistical output into an actionable, non-technical executive summary.

## Technical Requirements Implemented

- **Data Distributions:** Used `stats.shapiro()` to test for normality, mathematically proving the sales data is right-skewed.
- **Hypothesis Testing:** Implemented `stats.ttest_ind()` and `stats.f_oneway()` to evaluate demographic and regional variance against total sales and churn rates.
- **Regression Modeling:** Leveraged `statsmodels.api` to fit a linear regression model, calculating R-squared to define the exact impact of volume on revenue.
- **Confidence Intervals:** Deployed `stats.t.ppf()` and `stats.sem()` to establish a 95% confidence boundary for the population mean.

## Data Manipulation & Statistical Logic

The program utilizes the following advanced statistical logic to process and present the data:

- Normality Test - Evaluates if the data follows a bell curve. (Result: Non-normal, highly skewed by VIP purchases).
- Independent T-Test - Compares the average monthly charges of churned vs. retained customers to see if pricing drives churn.

- ANOVA Testing - Evaluates multiple categories simultaneously to see if geography significantly impacts sales averages.
- Confidence Interval - Calculates the exact margin of error, allowing the business to project minimum and maximum average revenue accurately.
- Linear Regression - Builds a mathematical model to quantify exactly how much total revenue variance is dictated purely by quantity sold.

## Setup & Installation

1. Install Python on your system
2. Install required libraries by running: ``pip install pandas numpy scipy statsmodels matplotlib seaborn jupyter``
3. Download or clone this repository
4. Ensure ``sales_data.csv`` and ``customer_churn.csv`` are placed in the root directory
5. Run the notebook using Jupyter: ``jupyter notebook statistical_analysis.ip``

## In-Depth Statistical Result Analysis

### Test 1: Customer Churn vs. Monthly Charges (Independent T-Test)

- Null Hypothesis: No difference in monthly charges between churned and retained customers.
- Result: p-value = 0.0163
- Analysis: Because  $p < 0.05$ , we reject the null hypothesis. The amount a customer is charged per month is a statistically significant driver of churn.

### Test 2: Regional Sales Variance (ANOVA)

- Null Hypothesis: Total sales averages are equal across all geographical regions.
- Result: p-value = 0.0972
- Analysis: Because  $p > 0.05$ , we fail to reject the null hypothesis. There is no mathematical proof that any specific region is outperforming or underperforming the others.

### Test 3: Linear Regression (Quantity vs. Total Sales)

- Result: R-squared = 0.4735 | p-value = 2.58e-15
- Analysis: Quantity sold accounts for 47.35% of the variance in Total Sales. The nearly zero p-value indicates this relationship is highly significant.

### Confidence Interval (Total Sales)

- Result: 123,650.48 INR +/- 19,874.13 INR
- Analysis: We are 95% confident that the true population average for transaction revenue falls between 103,776.35 INR and 143,524.61 INR.